Haseeb Javed. Ph.D.



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Haseeb Javed is Ph.D. graduate researcher experienced in designing and implementing Large Language Model (LLM) pipelines, including instruction fine-tuning, parameter-efficient adaptation (LoRA/PEFT), and domain-specific model optimization. Proficient in building Retrieval-Augmented Generation (RAG) frameworks with vector databases, embedding models, and hybrid retrieval strategies to enhance factual grounding and reduce hallucinations. Familiar with Agentic AI architectures involving tool use, multi-step reasoning, memory integration, and autonomous decision-making workflows. Applied GenAI methods in research-driven systems, emphasizing interpretability, efficiency, and scalable deployment.

EDUCATION

2022-Present **SUNGKYUNKWAN UNIVERSITY** **Suwon, KR**
PhD Computer Science and Engineering; CGPA: 4.25

• **Research Focus:** Developing computationally efficient AI systems with emphasis on Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Agentic AI architectures, optimizing model performance, scalability, and interpretability across real-world ML applications.

• **Technical Proficiency:** Advanced expertise in Python (PyTorch, TensorFlow, Scikit-learn), along with Java, C++, and R, for implementing end-to-end ML/LLM pipelines, fine-tuning (LoRA/PEFT), embedding-based retrieval systems, and scalable data processing workflows.

• **Applied AI & Modeling:** Designed dynamic ensemble frameworks, high-dimensional statistical models, and retrieval-integrated learning systems to extract structured insights from large-scale data, enhancing predictive accuracy, robustness, and decision intelligence.

2019-2021 **MUHAMMAD NAWAZ SHARIF UNIVERSITY OF ENGINEERING & TECHNOLOGY (UET MULTAN)** **Multan, PK**
Master of Project Management; CGPA: 3.791

- Specialized in Agile and project management methodologies, focusing on resource allocation optimization and risk mitigation strategies.

2020-2021 **MUHAMMAD NAWAZ SHARIF UNIVERSITY OF ENGINEERING & TECHNOLOGY (UET MULTAN)** **Multan, PK**
Master of Science in Electrical Engineering; CGPA: 3.613

- Expertise in designing energy management systems using ML algorithms for microgrids, incorporating demand-response strategies.

2014-2018 **UNIVERSITY OF ENGINEERING & TECHNOLOGY (UET TAXILA)** **Taxila, PK**
Bachelor of Science, in Electrical Engineering

PROFESSIONAL EXPERIENCE

Researcher
Sungkyunkwan University (SKKU), Suwon, South Korea
2022 – Present

- Leading research projects focusing on machine learning (ML) and deep learning (DL), with a specialization in model robustness and explainability.
- Designed and developed neural network models for early diagnosis of patient's disease using medical imaging and structured prediction techniques.
- Collaborated with interdisciplinary teams on algorithm optimization, resulting in published research in high-impact journals.

Teaching Assistant – Computing and Informatics Faculty
Sungkyunkwan University (SKKU), Suwon, South Korea
2023 – 2023

- Assisted in undergraduate and graduate-level courses related to computing and informatics, focusing on algorithms, data structures, and ML foundations.
- Provided mentorship to students on practical programming assignments, research methodologies, and project implementations.

Research Assistant
Muhammad Nawaz Sharif University of Engineering & Technology (UET Multan), Pakistan
2019 – 2021

- Conducted research under the supervision of HEC-approved supervisor Dr. Khawaja Abdul Muqeet, focusing on energy systems and AI integration.
- Published over six peer-reviewed articles in journals such as MDPI Energies (Impact Factor 3.004), Sensors (3.8), and Energy Sustainability & Society (2.811).
- Contributed to multi-disciplinary projects involving AI-based solutions for sustainable infrastructure and smart grid systems.

RESEARCH PROJECTS PUBLICATIONS

https://scholar.google.com/citations?user=Px3nMw0AAAAJ&hl=en **(Google Scholar)** **Published: Total Impact Factor: 40+**

Published Research

- Robustness in Deep Learning Models for Medical Diagnostics: Security and Adversarial Challenges Towards Robust AI Applications
- Impact of AI and Dynamic Ensemble Techniques in Enhancing Healthcare Services: Opportunities and Ethical Challenges
- Ethical Frameworks for Machine Learning in Sensitive Healthcare Applications
- An Ensemble Stacking Algorithm to Improve Model Accuracy in Bankruptcy Prediction
- A Comparative Analysis of Artificial Intelligence and Machine Learning Approach to Estimate Currents in Electrical Power Transmission Lines

Main Research Areas

• **Agentic AI Systems & Autonomous LLM Architectures:**
Design and development of agent-based AI systems capable of multi-step reasoning, tool integration, memory management, and adaptive decision-making. Research focuses on scalable orchestration frameworks, autonomous workflows, and reliability optimization for real-world AI deployment.

• **Retrieval-Augmented Generation (RAG) & Knowledge-Grounded LLMs:**
Development of advanced RAG pipelines integrating embedding models, vector databases, hybrid retrieval strategies, and factual grounding mechanisms to reduce hallucination and improve domain-specific accuracy in large language models.

• **Parameter-Efficient LLM Adaptation & Optimization:**
Research on LoRA/PEFT-based fine-tuning, domain adaptation, quantization (INT8/INT4), and computational efficiency techniques to enhance scalability and performance of generative AI systems.

• **Trustworthy & Interpretable Generative AI:**
Improving transparency, uncertainty estimation, and controllability in LLM-driven systems for high-stakes applications such as healthcare, cybersecurity, and enterprise AI.

PROFESSIONAL CERTIFICATIONS

Coursera.org/certifications/haseeb-javed **(Coursera Certifications)** coursera

Jan, 2022 **MACHINE LEARNING**
Standford University - 3 Months - (by Andrew NG)

Jan, 2022 **DEEP LEARNING (5 COURSES SPECIALIZATION)** **Coursera**
DeepLearning.AI - 4 Months - (by Andrew NG)
1. *Neural Networks and Deep Learning*
2. *Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization*
3. *Structuring Machine Learning Projects*
4. *Convolutional Neural Networks*
5. *Sequence Models*

Jan, 2022 **PYTHON FOR EVERYBODY (4 COURSES SPECIALIZATION)** **Coursera**
University of Michigan - 4 Months - (by Charles Severance)
1. *Programming for Everybody (Getting Started with Python)*
2. *Python Data Structures*
3. *Using Python to Access Web Data*
4. *Capstone: Retrieving, Processing, and Visualizing Data with Python*

Technical Skills

Programming & Development:
• Python (PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas)
• C++, Java, R
• MATLAB

Generative AI & Machine Learning:
• Large Language Models (LLMs), LoRA/PEFT Fine-Tuning
• Retrieval-Augmented Generation (RAG) Systems
• Agentic AI Architectures & Multi-Step Reasoning
• Dynamic Ensemble Learning & Statistical Modeling
• NLP, Topic Modeling, Text Classification

Data & Analytics Tools:
• Data Preprocessing, Feature Engineering, Model Evaluation
• Web Analytics & Data Visualization

Productivity & Documentation:
• Microsoft 365 (Word, Excel, PowerPoint)